TALHA ÇAĞLAR

AI Engineer Candidate | LLM Applications · Python · Open Source

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PROFILE

Final-year Computer Engineering student moving into AI engineering. Builds applications around structured LLM output, schema validation and multi-provider integration, and is taking AI engineering courses. Three internships across web security testing, enterprise software and IT infrastructure. Four open-source apps are packaged on the AUR.

AI ENGINEERING TRAINING

Building with the Claude API · *Anthropic Academy* — Oct 2026 – In progress

Completed: accessing Claude through the API, prompt evaluation (test datasets, code and model-based grading), prompt engineering. Remaining: tool use, RAG, prompt caching, MCP, agent and workflow patterns.

AI Engineer Study Program · *Self-directed, project-based* — Oct 2026 – Feb 2027 (Expected)

LLM APIs, RAG, agents and MCP, evaluation and deployment; each stage ends with a finished project. Next courses: Retrieval Augmented Generation (DeepLearning.AI), Introduction to MCP (Anthropic), AI Agents Course (Hugging Face).

AI PROJECTS

Akıllı Ekstre · *Python, FastAPI, Pydantic, OpenAI API, Next.js* — Private repository

- Kept the financial calculations of a bank statement analyzer in deterministic services and added an optional LLM explanation layer; the model writes no numbers or free text and only selects from a catalog of approved copy.
- Responses that fail Structured Outputs and Pydantic validation never reach the user or the cache. An allowlist keeps names, IBANs and transaction text away from the provider; imported text stays separate from instructions.

Short Video Automation · *Python, Gemini, Claude, OpenRouter, faster-whisper, FFmpeg* — Private repository

- Two pipelines: topic to voiced, captioned vertical video; long video to short clips via LLM selection over a transcript.
- Abstracted three LLM providers behind one layer; added structured JSON, caching, fact-checking and human approval.

ClarSec · *Python, Typer, Pydantic, httpx, pytest* — In development

- Terminal assistant that parses saved Nmap output deterministically and produces rule-based findings; optional LLM analysis passes typed JSON validation and exact evidence matching, and falls back to the rules on provider errors.

Lexis · *Python, PyQt6, Gemini, SQLite* — github.com/talhacaglar/Lexis · AUR: lexis-git

- Local-first dictionary app: optional Gemini assistance with Pydantic-schema JSON, backoff retries, SM-2 repetition.

Other projects: Notion Dictionary Automation (Gemini API); packaged on the AUR: Omarchy Focus, Pritingo, ceviribro.

EXPERIENCE

PTT KULE A.Ş. · *IT Intern · Çamlıca Tower, İstanbul* — Aug 2026 – Sep 2026

- Helped troubleshoot faults across the IT infrastructure; resolved connectivity issues for VLAN/PoE IP cameras.

İşbir Sentetik Dokuma Sanayi A.Ş. · *IT Intern · Balıkesir* — Jul 2026 – Aug 2026

- Built PrintHub, an Active Directory-integrated printer and toner management tool (Electron, Express, SQLite), with SNMP device discovery, LDAP access, role-based permissions and audit logs.

EMA Security LTD · *Cyber Security Intern (Red Team) · Remote* — Oct 2025 – Mar 2026

- Performed authorized web security testing and OWASP Top 10 vulnerability analysis; validated and reported findings.

EDUCATION

Sakarya University of Applied Sciences · *B.Sc. in Computer Engineering (4th year)* — Oct 2023 – Jun 2027 (Expected)

TECHNICAL SKILLS

LLM & AI	Claude, OpenAI, Gemini APIs; Structured Outputs, Pydantic, prompt engineering and evaluation
AI Tooling	Claude Code, Codex; hook, skill and MCP integrations; persistent-memory agent setup (ClarchOS)
Software	Python (FastAPI, PyQt6), TypeScript (Next.js), JavaScript (Electron), C++ (Qt), SQLite, pytest, Git
Linux & Network	Arch Linux, Bash, AUR packaging, TCP/IP, VLAN, DNS/DHCP, SNMP, LDAP/Active Directory
Security	OWASP Top 10, web application testing, vulnerability analysis · Burp Suite, Nmap, Wireshark
Languages	Turkish (native), English (B2)

CERTIFICATIONS & ACHIEVEMENTS

Deep Learning for Image Processing · Dijital Genç AI Ecosystem · Achievement & Participation · exam 95/100 — Oct 2024

Machine Learning · Dijital Genç AI Ecosystem · Achievement & Participation · exam 85/100 — Oct 2024

TEKNOFEST 2024 Robolig Competition (University level) — Finalist · Team AllSpark · interface design — Oct 2024

CWSE — Certified Web Security Expert · Hackviser · HV-CWSE-ZNKC7QAY — Feb 2026

CAPT — Certified Associate Penetration Tester · Hackviser · HV-CAPT-2U0W5XSE — Oct 2025

Türkiye Siber Vatan Program 2026 · Certificate of Participation — Mar 2026